



Lab 07: Setting up a Private Kubernetes Cluster

Cloud Security Fundamentals | GSP178

Overview

Created private Kubernetes Engine clusters where nodes have no public IP addresses.
Configured master authorized networks to control which IPs can access the cluster master.
Deployed both auto-subnetwork and custom subnetwork private clusters.

Private vs Public Kubernetes Cluster

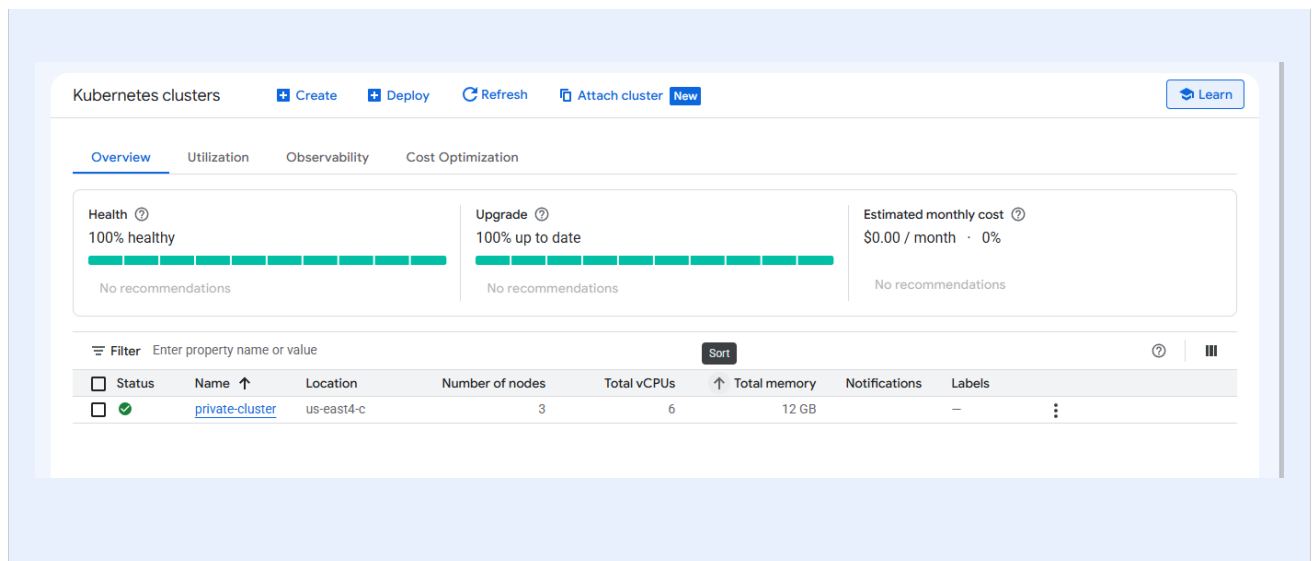
Private Cluster	Public Cluster
Nodes have no public IPs	Nodes have public IPs
Master accessible via authorized networks only	Master accessible from public internet
Workloads isolated from internet	Workloads potentially exposed

Task 1 - Set Region and Zone

```
gcloud config set compute/zone [ZONE]
export REGION=[REGION]
export ZONE=[ZONE]
```

Task 2 - Create a Private Cluster

```
gcloud beta container clusters create private-cluster --enable-private-nodes --
master-ipv4-cidr 172.16.0.16/28 --enable-ip-alias --create-subnetwork "" --machine-
type e2-medium
```



Task 3 - View Subnet and Secondary Ranges

```
gcloud compute networks subnets list --network default
gcloud compute networks subnets describe [SUBNET_NAME] --region=$REGION
```

Output shows primary range for nodes and two secondary ranges for pods and services.

Task 4 - Enable Master Authorized Networks

```
gcloud compute instances create source-instance --zone=$ZONE --machine-type=e2-medium --scopes 'https://www.googleapis.com/auth/cloud-platform'
gcloud compute instances describe source-instance --zone=$ZONE | grep natIP
gcloud container clusters update private-cluster --enable-master-authorized-networks --master-authorized-networks [MY_EXTERNAL_RANGE]
```

SSH'd into source-instance, installed kubectl, and verified nodes have no external IPs.

```
kubectl get nodes --output yaml | grep -A4 addresses
kubectl get nodes --output wide
```

```
subnetwork entry generated for private-cluster:
student-01-6c5bb1f48c8e@source-instance:~$ kubectl get nodes --output wide
NAME                                STATUS    ROLES    AGE   VERSION           INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                                     KERNEL-VERSION   CONTAINER-RUNTIME
gke-private-cluster-default-pool-ca7f0e8f-33qd   Ready    <none>    11m   v1.35.3-gke-2190000  10.125.4.3    <none>        Container-Optimized OS from Google         6.12.68+         containerd://2.1.7
gke-private-cluster-default-pool-ca7f0e8f-8qzw   Ready    <none>    11m   v1.35.3-gke-2190000  10.125.4.2    <none>        Container-Optimized OS from Google         6.12.68+         containerd://2.1.7
gke-private-cluster-default-pool-ca7f0e8f-vtxd   Ready    <none>    11m   v1.35.3-gke-2190000  10.125.4.4    <none>        Container-Optimized OS from Google         6.12.68+         containerd://2.1.7
```

Task 5 - Clean Up

```
gcloud container clusters delete private-cluster --zone=$ZONE
```

Task 6 - Create Private Cluster with Custom Subnetwork

```
gcloud compute networks subnets create my-subnet --network default --range 10.0.4.0/22 --enable-private-ip-google-access --region=$REGION --secondary-range my-svc-range=10.0.32.0/20,my-pod-range=10.4.0.0/14
gcloud beta container clusters create private-cluster2 --enable-private-nodes --enable-ip-alias --master-ipv4-cidr 172.16.0.32/28 --subnetwork my-subnet --services-secondary-range-name my-svc-range --cluster-secondary-range-name my-pod-range --zone=$ZONE --machine-type e2-medium
```

Gemini CLI is available in Cloud Shell terminal! Type `gemin1` to try it. [Learn more](#) Don't show again

```
Deleted [https://container.googleapis.com/v1/projects/qwiklabs-gcp-02-3d50c1568b61/zones/us-east4-c/clusters/private-cluster].
student_01_6c5bb1f48c8e@cloudshell:~ (qwiklabs-gcp-02-3d50c1568b61) $ gcloud compute networks subnets create my-subnet \
--network default \
--range 10.0.4.0/22 \
--enable-private-ip-google-access \
--region=$REGION \
--secondary-range my-svc-range=10.0.32.0/20,my-pod-range=10.4.0.0/14
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-02-3d50c1568b61/regions/us-east4/subnetworks/my-subnet].
NAME: my-subnet
REGION: us-east4
NETWORK: default
RANGE: 10.0.4.0/22
STACK TYPE: IPV4 ONLY
IPV6 ACCESS TYPE:
INTERNAL IPV6 PREFIX:
EXTERNAL IPV6 PREFIX:
student_01_6c5bb1f48c8e@cloudshell:~ (qwiklabs-gcp-02-3d50c1568b61) $ gcloud beta container clusters create private-cluster2 \
--enable-private-nodes \
--enable-ip-alias \
--master-ipv4-cidr 172.16.0.32/28 \
--subnetwork my-subnet \
--services-secondary-range-name my-svc-range \
--cluster-secondary-range-name my-pod-range \
--zone=$ZONE \
--machine-type e2-medium
Creating cluster private-cluster2 in us-east4-c... Cluster is being health-checked (Kubernetes Control Plane is healthy)...done.
Created [https://container.googleapis.com/v1beta1/projects/qwiklabs-gcp-02-3d50c1568b61/zones/us-east4-c/clusters/private-cluster2].
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload/_gcloud/us-east4-c/private-cluster2?project=qwiklabs-gcp-02-3d50c1568b61
kubeconfig entry generated for private-cluster2.
NAME: private-cluster2
LOCATION: us-east4-c
MASTER VERSION: 1.35.3-gke.2190000
MASTER IP: 34.85.160.164
MACHINE TYPE: e2-medium
NODE VERSION: 1.35.3-gke.2190000
NUM NODES: 3
STATUS: RUNNING
STACK TYPE: IPV4
student_01_6c5bb1f48c8e@cloudshell:~ (qwiklabs-gcp-02-3d50c1568b61) $ gcloud compute instances describe source-instance --zone=$ZONE | grep natIP
```

What I Learned

- Private clusters isolate workloads from the internet by removing public IPs from nodes
- Master authorized networks restrict which IPs can manage the cluster via kubectl
- IP aliases allow Kubernetes to manage pod and service IP ranges separately from nodes
- A /28 CIDR range is required for the master VMs
- privateIPGoogleAccess allows nodes without public IPs to reach Google APIs
- Custom subnets give full control over pod and service IP ranges

Resources

GCP Private Clusters Docs: <https://cloud.google.com/kubernetes-engine/docs/how-to/private-clusters>

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